6/27

APTITUDE

Day - 4 (Time & work)

1) A can complete a work in 10 days and B in 15 days. How long will they take together.

$A \rightarrow 1/10$

$B \Rightarrow \frac{1}{15}$

$A + B \Rightarrow 1/10 + 1/15$

we take LCM

$\Rightarrow 1^{\times 3}/10\times 3 + 1/15^{\times 2} \times 2$

$\Rightarrow \frac{3+2}{30}$

$\Rightarrow \cancel{5/30}\ 6$

$\boxed{\Rightarrow 1/6}$ Answer.

2) A, B and C can complete a work in 6, 8 and 12 day respectively. How many days will they take working together?

$A \rightarrow 1/6$

$B \rightarrow 1/8$

$C \rightarrow 1/12$

$A + B + C = 1/6\times 4 + 1/8\times 3 + 1/12 \times 2$

We take LCM

$\Rightarrow \frac{4+3+2}{24}$

$\Rightarrow \cancel{9/24}\ 8 \rightarrow 3/8$

Reciprocal $\rightarrow 8/3 \rightarrow 2\frac{1}{3}$ days.

3. A alone can do a work in 12 days. After 4 days, B joins and completes remaining work in 4 days. Find Time taken by B alone.

$A \rightarrow 1/12$

4 days

$\Rightarrow 4 \times (1/12)$

$\Rightarrow 1/3$

Remaining work $= 1 - 1/3$

$= \frac{1-3}{3}$

$\boxed{\Rightarrow 2/3}$

$(A+B) \Rightarrow (2/3) \div 4$

$= 2/12$

$\Rightarrow 1/6$

$\boxed{A = 1/12}$

$\frac{1 \times 2}{6 \times 2} - 1/12$

$= \frac{2-1}{12}$

B alone $\rightarrow \boxed{= 1/12}$ Answer.

4. If 5 workers can complete a work in 8 days, how many workers are required to complete it in 4 days?

5 workers $\rightarrow$ 8 days

Total work $\rightarrow 5 \times 8 \rightarrow 40$ units

For 4 days:

workers $= 40 \div 4 \rightarrow \boxed{10}$

$\boxed{\text{10 workers}}$ Answer

5. A pipe fills a tank in 6 hours and another empties it in 8 hours. How long to fill the tank?

$A \Rightarrow 1/6$

$B \Rightarrow 1/8$

$\Rightarrow \frac{1 \times 4}{6 \times 4} - \frac{1 \times 3}{8 \times 3}$ ⇒ we take LCM

$\Rightarrow \frac{4-3}{24}$

$\boxed{\Rightarrow 1/24}$ ⇒ Answer

Block 2 - VERBAL ABILITY

Reading Passage

1) why are optimized systems important

optimized systems important because they improve efficiency, speed and performance.

2) what does scalability mean in real-world applications?

Scalability means a system can handle the data and users efficiently.

3) Name companies that depend on scalable systems.

* Amazon
* Google

Grammar Correction:

1) He didn't knew the answer.

He didn't know the answer.

2. She is more smarter than him.

She is smarter than him

3. They has completed the task.

They have completed the task.

Block - 3

COMMUNICATION

Arrays.

Array is a collection of similar data items.

It stores the elements in continuous memory locations.

Real-world Eg:

Student Marks, Shopping list

Uses case: online voting system

votes = {120, 95, 150}

⇒ Each index represents a candidate

⇒ It is very useful to find winners easily.

DBMS. → Database Management system

DBMS used to store and manage the data efficiently. It helps to organizing large amount of data in structure way.

Real world Eg: Bank System.

⇒ In bank system, all customer details like account number, balance, transactions are stored in a Database.

DBMS helps to quickly access and update the data.

USE CASE.

=> Banking systems -> to store customer data

=> College Systems -> to manage student records.

OS

Operating System is system software that manages computer hardware and software resources and act as an Interface between the user and computer.

Real-time Example.

ATM Machine

when we ~~enter~~ use an ATM: like we insert ATM card, Enter pin and withdraw money. -> It process you input, connect to the bank server and display options on screen.

USE CASE:

Multitasking

Resource Management

- CPU (processing
- Memory
- Storage.

File Management